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Identifying Risk Sources with Principal Component Analysis

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Abstract

Risk is an important characteristic that plays a huge role in capital markets decision making. Thus, correctly determining and specifying the risk sources on a given market may shape investors' decisions. This thesis aims to identify and interpret the sources of financial risk on the German equity market. The goal is achieved using principal component analysis as a tool for building the portfolios that allocate their budgets to distinct uncorrelated risk sources. Throughout the research, correlation matrix was compared with the covariance matrix as an underlying foundation for PCA, effectively examining if standardising the returns is an important step. It was also revealed that the first five principal components have different interpretations, indicating the underlying equity characteristics that may not be obvious on a first sight, effectively revealing the biggest risk sources on the German equity market. The market component was extracted that accounted for over 30% of the overall variance, representing a big portion of systematic risk. The findings of the study also suggest that normalising the returns might be a beneficial approach.

Keywords

Principal component analysis, Financial risk, Equity markets

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Introduction

In the dynamic landscape of global finance, capital markets stand as the bedrock upon which economics flourish, investments thrive, and innovation is fueled. The intricacies of capital markets, comprising stocks, bonds, commodities, and derivatives, wield significant influence over the allocation of resources, the valuation of assets, and determination of risk and return profiles. As the focal point of myriad of economic activities, understanding the mechanisms, behaviours, and interdependencies within capital markets is not only crucial for investors and financial institutions but also holds profound implications for policymakers, regulators, and society at large.

Risk is an important characteristic that plays a huge role in asset markets decision making. It shapes investors' strategies, portfolio allocations, and overall approach to market participation. Understanding and effectively managing risk is paramount for mitigating potential losses and optimising the returns. Consequently, identifying the risk sources for proper risk budget allocation can constitute an integral component for one's success on a capital market. The central concept here revolves around spreading your capital across multiple sources, a strategy commonly referred to as diversification.

Even though the idea of diversification was known for as long as the beginning of 20th century, (Markowitz, 1952) popularised the said concept and introduced the idea of overall portfolio risk and return being taken into account instead of those of singular securities. These ideas helped quantifying the portfolio selection using statistical properties of combined assets' returns, such as mean, variance and covariance. The general aim of modern portfolio theory is to have diversified assets portfolios that have maximised expected return for a given financial risk exposure. The theory makes an assumption that average investor should be risk-averse and give preferences to less risky portfolios given the utility function $U = f(E_R, \sigma_R)$.

(Sharpe, 1964) followed up on the said ideas by introducing the Capital Asset Pricing Model, where he constructed a market equilibrium theory of asset prices under the condition of risk. It was observed that the risk of a portfolio consists of two components: systematic (undiversifiable) and idiosyncratic risks. Systematic risk refers to the risk that all assets share, in other words, market risk, which is caused by common economic activity. It was proven that diversification enables the investor to escape all but the systematic exposure.

In practice, most financial portfolios consist of multitude of various assets and their yields pivots on many external and internal variables. Since the rise of computational power in the mid-late 20th century and the abundance of available economic and financial data, factor models were emerging, albeit known since the works of (Spearman, 1904). Following (Connor, 1995) classification, one may explore three types of financial factor models: macroeconomic, which take into account such factors as inflation rate, unemployment and etc.; fundamental, which build up their explanation of returns based on various attributes internal to assets: firm size, dividend yield, book-to-market ratio etc.; the last are the statistical models, which

use numerous maximum-likelihood and principal-components-based approaches on time series samples of asset returns to identify the pervasive factors.

(Partovi and Caputo, 2004) delved into the idea of reformulating the original efficient portfolio problem introduced half a century ago by (Markowitz, 1952) by reorganising the correlated assets set into an uncorrelated mixture of assets with the help of principal component analysis, thus coining the term “principal portfolio”. This greatly simplifies the general problem of portfolio optimisation. It was also highlighted that with the allowance of short trades, it is always possible to reorganise the entire asset set as an equivalent set of principal portfolios and that the elimination of correlations is emphasized as the key factor in achieving a major reduction of price volatility, simultaneously hedging and leveraging the principal portfolios.

In this thesis the concept of statistical models will be explored further. In particular, the application of Principal Component Analysis (PCA) as a tool for identifying uncorrelated risk sources on German equity market. The said analysis was done by many academics and practitioners alike, in particular one can note the works of (Avellaneda and Lee, 2010), (Lohre et al., 2012) and (Avellaneda, Healy, et al., 2020) that partially conducted similar type of work for the U.S equity market.

The thesis will be structured as following: firstly, the linear algebra apparatus used in PCA will be demonstrated as well as the data gathering and preprocessing will be explained and the exploratory data analysis will be conducted to derive insights into the equities. Then, the principal components will be constructed on covariance and correlation matrices to investigate the differences between two approaches and what could be the probable cause of them. Lastly, the possible explanation of principal components using the eigenvector and regression analysis will be derived.

Research questions

The goal of this thesis is to answer two questions:

1. What is the difference between using correlation and covariance matrix as an underlying foundation for PCA in the context of financial markets? In other words, is it necessary to standardise the data before applying the PCA?
2. What can be the possible interpretations of the first N principal components? How are they different?

1. Methodology

The methodology chapter of this thesis provides a detailed overview of the research design, data collection methods, and analytical techniques employed to investigate how the risk sources can be identified using principal component analysis. Through a comprehensive exposition of the research methodology, readers are provided with insights into the methodological rigour and integrity underpinning the study, ensuring transparency and reproducibility.

1.1 Data gathering and preprocessing

As was mentioned previously, the German equity market was chosen as an example of applying the PCA to extract principal portfolios as opposed to a textbook example of the US equity market, because of numerous reasons. Firstly, Germany's economic structure and regulatory environment differ significantly from the US's. Secondly, a unique corporate landscape, characterised by a strong emphasis on manufacturing and export-oriented industries, whereas the US's is more contrast and diverse, which can be attested to the government's policies. Thirdly, Germany's market operates within the framework of the Eurozone, subject to its monetary policies, currency fluctuations, and broader economic integration efforts. Lastly, even in the era of globalisation, there are cultural and institutional differences, such as corporate governance practices, investor preferences, that contribute to market behaviours and investment strategies. All of these points make the German equity market quite an interesting case study, which will be explored further.

Since the asset universe is enormous in the case of a whole market, it was decided to construct a proxy for it in a similar manner as in (Avellaneda and Lee, 2010). Generally, to capture the state of a market, it would be sufficient to have companies of different sizes, coming from different sectors and industries to provide a heterogeneous sample. With these points in mind, it was determined to choose constituents of three indices: DAX, MDAX and SDAX. The first one consists of 40 major German blue chip companies, the second one contains 50 Prime Standard enterprises, that are right below the DAX index size-wise, the third one falls right after the MDAX index in size. The total number of stocks accumulates to 160.

The sample data time window was chosen to start at the beginning of 2010, marking the start of a new decade, and end on 31th of August, 2023. The data itself was downloaded from publicly available Yahoo! Finance API resource, all of the equities are listed on Frankfurt Stock Exchange. Only the companies that existed throughout the whole time period were retained, thus the number of equities was shrunk to 107. In case if equities missed some entries, the latest available data was used to propagate its value further.

Lastly, the price series were converted to returns series to induce stationarity using the

following formula from (Tsay, 2010):

$$R_t = \frac{P_t}{P_{t-1}} - 1, \quad (1.1)$$

where R_t is a result return or a price change at time t and P_t is a price of an asset at time t . (Tsay, 2010) also states that these simple returns are assumed to be independently and identically distributed as normal with fixed mean and variance.

1.2 Statistical tests

In this section, various statistical tests that will be used in this work will be thoroughly described, e.g. the test statistics and their distribution will be stated, as well as the motivation behind these tests will be provided.

1.2.1 Jarque and Bera test

In their work, (Jarque and Bera, 1987) derive a statistical approach for testing for normality, that has good finite sample performance, which states:

H_0 : the distribution of a given time series follows normal

H_1 : the distribution of a given time series does not follow normal

The test statistic is following:

$$JB = \frac{\hat{S}^2(R)}{6/T} + \frac{(\hat{K}(R) - 3)^2}{24/T}, \quad (1.2)$$

where $\hat{S}^2(R)$ is sample skewness of a given return series, $\hat{K}(R)$ is sample kurtosis and T is the number of observations in a given series. This test statistics is asymptotically distributed as a chi-squared random variable with 2 degrees of freedom. The test basically compares both skewness and kurtosis of a given random variable with that of normal distribution via t-ratio, than sums both statistics up and tests it against χ^2 distribution, which is a transformed version of normal distribution.

1.2.2 Autocorrelation tests

According to (Tsay, 2010), besides usual correlation between two variables, there also exists a phenomenon of autocorrelation, when the variable is said to be correlated with its past values. This is particularly resemblant in economical and financial time series, since following value usually succeeds its predecessor and some shocks that happened in past may still have an effect on subsequent observations. The concept is widely used in the linear time series analysis context, mainly in the concept of autoregressive models (AR) and its various extensions.

There are multiple ways of testing, whether a given time series is autocorrelated, such as an autocorrelation function (ACF) which is defined as following:

$$\rho_\ell = \frac{Cov(r_t, r_{t-\ell})}{Var(r_t)}. \quad (1.3)$$

However, as it only tests for one particular lag, it can be proven to be quite a tedious task to test for multiple lags of one time series, let alone for a whole asset portfolio. That's the reason why Portmanteau tests emerged: effectively, they test for the presence of joint autocorrelation up to a particular lag.

(Ljung and Box, 1978) built up on work of (Box and Pierce, 1970) and introduced a test for autocorrelation of a given random variable, which has following hypotheses:

H_0 : the data is independently distributed, e.g. the autocorrelations are not statistically significant and any observed correlations in the data result from randomness

H_1 : the data is not independently distributed, it exhibits serial correlation

The test statistic is:

$$Q(m) = T(T+2) \sum_{\ell=1}^m \frac{\hat{\rho}_\ell^2}{T-\ell}, \quad (1.4)$$

where T is the number of observations in a given series, m is the number of lags and $\hat{\rho}_\ell$ is the sample autocorrelation at lag ℓ . The decision rule is to reject H_0 if $Q(m) > \chi_\alpha^2$.

(Tsay, 2010) suggests that the number of lags m should be chosen by the relationship $m \approx \ln(T)$.

1.2.3 Testing for unit root

According to (Tsay, 2010), one of the most important prerequisites for applying a Principal Component Analysis is for the time series to be stationary. There exists two types of this phenomenon: strict stationarity and weak stationarity. The first one requires that the joint distribution of $(R_{t_1}, \dots, R_{t_k})$ is invariant under time shift, which is a very strong condition that is very hard to verify empirically. A time series $\{R_t\}$ is said to be *weakly stationary* if both $E(R_t) = \mu$ and $Cov(R_t, R_{t-\ell}) = \gamma_\ell$ are true. One can notice that both statements are time-invariant, which ensures weakly stationarity. (Dickey and Fuller, 1979) have developed a test for unit-root, that employs the model:

$$p_t = \phi_0 + \phi_1 p_{t-1} + e_t, \quad (1.5)$$

where p_t is a given time series at time t and e_t denotes the error term. The hypotheses for this test are:

H_0 : $\phi_1 = 1$

H_1 : $\phi_1 < 1$

The test statistic is a t -ratio of the least squares estimate of ϕ_1 :

$$\frac{\hat{\phi}_1 - 1}{std(\hat{\phi}_1)}. \quad (1.6)$$

1.3 Theory of Principal Component Analysis

Applying the Principal Component Analysis on time series data can prove to be a tedious task, since the observations are not independent, which is induced by their proximity in time. Additionally, it can be further perturbed by seasonal dependencies underlying in the data. (Tsay, 2010) describes PCA as following: given a k -dimensional random variable $R = (R_1, \dots, R_k)'$ with covariance matrix Σ_R , a principal component analysis is concerned with using a *few* linear combinations of R_i to explain the structure of Σ_R , so it can be used to study the sources of variations of the asset returns.

PCA can be applied to either the covariance matrix Σ_R or the correlation matrix ρ_R of R , albeit the correlation matrix can be proven to be the covariance matrix of the standardised random vector $R^* = S^{-1}R$, where S is the diagonal matrix of standard deviations of the components of R . Covariance matrix will be used as an exemplary foundation of principal components in further discussion.

Let $w_i = (w_{i1}, \dots, w_{ik})'$ be a k -dimensional vector, where $i = 1, \dots, k$. Then

$$y_i = w_i' r = \sum_{j=1}^k w_{ij} r_j \quad (1.7)$$

is a linear combination of the random vector R . If R consists of the simple returns of k stocks, then y_i is the return of a portfolio that assigns weight w_{ij} to the j th stock. Since multiplying a constant to w_i does not affect the proportion of allocation assigned to the j th stock, the vector w_i can be standardised so that $w_i' w_i = \sum_{j=1}^k w_{ij}^2 = 1$. Then

$$\begin{aligned} Var(y_i) &= w_i' \Sigma_R w_i, \quad i = 1, \dots, k, \\ Cov(y_i, y_j) &= w_i' \Sigma_R w_j, \quad i, j = 1, \dots, k. \end{aligned} \quad (1.8)$$

The whole concept of PCA is to find linear combinations w_i such that y_i and y_j are uncorrelated for $i \neq j$ and the variances of y_i are as large as possible subject to constraint $w_i' w_i = 1$. Let $(\lambda_1, e_1), \dots, (\lambda_k, e_k)$ be the eigenvalue-eigenvector pairs of Σ_R , where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k \geq 0$. Following results will be derived: The i th principal component of R is $y_i = e_i' R$ for $i = 1, \dots, k$. Moreover,

$$\begin{aligned} Var(y_i) &= e_i' \Sigma_R e_i = \lambda_i, \quad i = 1, \dots, k, \\ Cov(y_i, y_j) &= e_i' \Sigma_R e_j = 0, \quad i \neq j. \end{aligned} \quad (1.9)$$

In addition,

$$\sum_{i=1}^k Var(R_i) = tr(\Sigma_R) = \sum_{i=1}^k \lambda_i = \sum_{i=1}^k Var(y_i). \quad (1.10)$$

Consequently, the proportion of total variance in R explained by the i th principal component is simply the ratio between the i th eigenvalue and the sum of all eigenvalues of Σ_R .

In practice, since the covariance or correlation matrix of the return series is unknown, the sample counterparts should be used as an estimation given that the returns are weakly stationary.

The number of principal components to retain can be estimated by examining the *scree plot*, which is the line plot of eigenvalues $\hat{\lambda}_i$ ordered from the largest to the smallest. One can observe an *elbow* at the graph, which indicates further steeping of the eigenvalues, e.g. the next principal component added will not explain as much variance compared to the previous ones.

1.4 Principal Portfolios

Of particular interest for this work is a concept of principal portfolios, which was coined in the works of (Partovi and Caputo, 2004). The said idea bears multiple names, one of which is eigenportfolio, which was used by (Avellaneda and Lee, 2010). As was mentioned in the Introduction, the original concept was to simplify the structure of the efficient portfolio by introducing the uncorrelated portfolios via the principal component analysis, which would effectively function as a new asset set. The authors mention that one of the constraints that makes the whole idea possible is the allowance of short trades, i.e. investing in such a way that one will profit if the value of the asset fails. It is also worth noticing that the result principal portfolios are leveraged and hedged. Finally, an important property that the authors emphasise is that the variance of a typical principal portfolio is about the same as that of a single asset in the original set. (Partovi and Caputo, 2004) also state that every investment environment $\{s_i, R_i, \sigma_{ij}\}_{i,j=1}^N$ which allows short sales can be recast as a principal portfolio environment $\{S_\mu, R_\mu, V_{\mu\nu}\}_{\mu,\nu=1}^N$. The principal portfolios are defined as following:

$$S_\mu = \sum_{i=1}^N \frac{e_i^\mu s_i}{W_\mu}, \quad (1.11)$$

where S_μ is the μ -th principal portfolio, s_i is the i -th asset and W_μ is defined as $W_\mu = \sum_{i=1}^N e_i^\mu$, effectively reflecting the relative investment weight of the μ -th eigenvector. It is important to acknowledge that the i -th element of the eigenvector e^μ may be referred to as an asset loading, since it represents how much of a weight does this particular eigenvector put onto one of the assets. An important note is that the eigenvectors should be chosen so that $W_\mu \geq 0$, where $W_\mu = 0$ implies that the portfolio is totally leveraged, i.e. the long and short trades' weights are equal. The authors also prove that the upper limit for W_μ is $N^{\frac{1}{2}}$, thus setting both lower and upper limits for it: $0 \leq W_\mu \leq N^{\frac{1}{2}}$, where the upper limit is reachable only for uniformly allocated portfolio. The expected rate of return and variance for each principal

portfolio is defined as

$$\begin{aligned} R_\mu &= \sum_{i=1}^N \frac{e_i^\mu R_i}{W_\mu}, \\ V_\mu^2 &= \sum_{j=1}^N \sum_{i=1}^N \frac{e_i^\mu \sigma_{ij} e_j^\mu}{W_\mu^2}. \end{aligned} \tag{1.12}$$

The equations 1.12 are to be referred as *expected principal rate of return* and *principal variance*, respectively. (Avellaneda and Lee, 2010) propose two more ways how to choose a number of principal components to retain: (a) take into account a fixed number of eigenvalues to extract the factors (assuming a number close to the number of industry sectors) or (b) take a variable number of components, in such a way that the truncation explains a given percentage of the total variance. In either case one will arrive at an array of significant eigenvalues $\lambda_1, \dots, \lambda_m, m < N$.

It is important to note that the principal component analysis is used for constructing the eigenportfolios as opposed to factor analysis because it has multiple advantages that benefit the whole concept according to (Tsay, 2010). Firstly, PCA produces orthogonal (uncorrelated) components, whereas factor analysis typically does not. The orthogonality is beneficial for principal portfolios, because it allows for each portfolio to represent a different risk source. Thus if an investor decides to allocate their risk budgets to multiple eigenportfolios, they will not be exposed to one source of volatility multiple times. Secondly, PCA aims to maximise the variance captured by each principal component. This property is desirable because it ensures that the resulting portfolios capture as much volatility as possible. Lastly, PCA provides a simpler and more transparent framework for constructing eigenportfolios, since the resulting principal portfolios are liner combinations of the original variables and are uncorrelated, the result is easier to interpret and implement in practice. However, it should be emphasised that the factor analysis approach also has its application in financial time series as a tool for factor investing, which is out of scope for this thesis.

1.4.1 Diversification

As was mentioned previously, one of the most crucial concepts during any financial portfolio assessment is the concept of diversification across different industries, sectors or other multiple criteria. This concept enables the investor to reduce risk and optimise returns. By diversifying their portfolios, investors aim to minimise the impact of adverse events that may affect any single investment or segment of a given market, as losses in one area may be offset by gains in another. Additionally, diversification can enhance potential returns by capturing opportunities across various dimensions. While diversification cannot eliminate all investment risk, it remains a crucial strategy for building resilient portfolios.

Continuing from previous discussion, one may remember that the principal component analysis will be managing the portfolio creation by effectively distributing the respective weights throughout the asset set. Two approaches will be compared: using a correlation and a covariance matrix of the given assets for constructing principal components. As the assessment

will be done not only from a statistical viewpoint, but also from a financial perspective, the performance of given portfolios has to be assessed as well. However, one needs to quantify the differences, that's why it was decided to refer to the work of (Choueifaty and Coignard, 2008) in order to use following equation:

$$DR(w) = \frac{w' \sigma}{\sqrt{w' \Sigma w}}, \quad (1.13)$$

where Σ is the $N \times N$ covariance matrix of the returns of the original asset universe, σ is the vector of asset volatilities measured by their respective standard deviations, w is a weight vector of a given portfolio. The higher the acquired result, the better the given portfolio in terms of diversification. Given equation will be applied to every single portfolio for both approaches, and then their metrics will be compared.

1.5 Fama-French Factors

For the sake of interpretation of the principal components, as was stated in one of the research questions, one can make use of already existing factors that were created using multiple financial fundamentals. An example of which was introduced by (Fama and French, 1993). In their groundbreaking work, the authors proposed common risk factors in the stocks' returns. These factors play a crucial role in the development of understanding the drivers of stock returns beyond the traditional Capital Asset Pricing Model. The original 3 factors that were introduced are Market Risk Factor ($Mkt - RF$), Size Factor ($SMB - Small minus Big$) and Value Factor ($HML - High minus Low$). The Market Risk Factor represents the excess return of the overall market portfolio over the risk-free rate, which was done by effectively incorporating market risk as a factor influencing equity returns. The SMB factor represents the difference in returns between the nine small stock portfolios and the nine big stock portfolios, where the size is measured by the equity's respective market capitalisation. The last of the original ones, the HML factors is the difference in average returns between the two value portfolios and two growth portfolios, where the respective metrics are being measured using the book-to-market ratio. High BE/ME ratio indicates a value stock and a low ratio - a growth stock. Later in (Fama and French, 2015), the authors have introduced two more factors, effectively composing a five factor model. The factors in consideration are the Profitability Factor ($RMW - Robust minus Weak$) and the Investment Factor ($CMA - Conservative minus Aggressive$). The first one is a difference between the average returns on the two robust operating profitability portfolios and the two weak operating portfolios. The last factor is the difference between two conservative investment portfolios and two aggressive ones. All the aforementioned factors were preconstructed for the European market by Kenneth French and provided on his website. The markets that were taken into consideration are the ones from the Western Europe and Northern Europe plus some Eurozone countries.

Originally, (Fama and French, 1993) proposed the factors mentioned above based on empirical observations from historical stock returns. The idea of their research was to challenge the conventional wisdom of the CAPM, where it was suggested that only systematic (market)

risk influences expected returns. Instead, it was argued that other factors might also play a significant role in explaining the variance in equity returns. In the scope of this thesis, the resulting principal portfolios will be compared with different factors to find a possible interpretation of the resulting portfolios. The particular method that will be used for comparison in this thesis is the regression analysis, that will make use of following linear relationship:

$$\mathbf{R}_{PP,i} = \mathbf{R}_{FF}\mathbf{B} + \epsilon, \quad (1.14)$$

where $\mathbf{R}_{PP,i}$ is the return series of the i -th principal portfolio, $\mathbf{R}_{FF}$ is the matrix of the returns of an arbitrary number of Fama-French components and B is the matrix of regression coefficients. The said equation was used by a number of practitioners to explain given principal portfolios, most notably, (Lohre et al., 2012).

2. Results

In this chapter the theoretical foundation will be used to conduct a case study for the German equity market. The results then will be presented in a systematic manner. The data collected during the research process will be analysed, thereby addressing the research questions posited in the study.

2.1 Exploratory Data Analysis

After acquiring the data, following the steps described in the section 1.1, a data set of size 3451×146 , was collected. However, it needs some preprocessing steps before it can be used for modelling. Firstly, the missing values were examined. Not all of the companies present in the indices at the point of writing were established back in the 2010, that accommodates for 38 companies that have more than 1% of data missing. Those stocks were decided to be discarded, since the missing values can not be meaningfully imputed. Surprisingly, even after this quite strict initial cleaning, some stocks are left with missing values. After careful examination of the said stocks, the missing values seem like a systematic error on the provider's side, since a pattern can be established. Noticeably, a big portion of stocks misses data on the same four days. Most probably, it was a communication error between the data provider and the capital market. Since the missing values are of system errors' origin, it was decided to fill them with the latest available price to the date of missing value, effectively propagating already known value forward. By doing this, it is avoided to look into the future from the standpoint of a missing value. After following these preprocessing steps, the price data is converted to the price change data R_t , following the formula 1.1. By doing this, the data was standardised. After examining the time series sample moments, following conclusions were drawn: means are generally positioned around 0; standard deviations usually lay in range between 0.012 and 0.036; One of the way to analyse sample means and standard deviations of the asset universe further is to plot them against each other.

The scatter graph 2.1 shows the distribution of stocks by average return on the y-axis and its volatility on the x-axis. Each dot represents an equity, the name is stated next to each dot. After examining the graph 2.1, one can divide the asset universe into four quadrants, starting from top left, counter clockwise: high return - high risk, high return - low risk, low return - low risk, low return - high risk. According to (Cipra, 2015), the direction of preference for a rational investor should be from the centre to the top right, since it maximises the return and minimises the risk. However, one can notice that this particular quadrant is emptier than the others in the given sample period. Most of the equities are grouped in the bottom left quadrant. Nonetheless, there are some noteworthy outliers present in the graphs, in particular VBK.DE, which is an example of high risk - high return stock, or SRT3.DE, which can be a better choice for a more rational investor.

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Skewness has some particularly interesting examples: 1&1 AG has a skewness of -2.72 , which indicates a distribution that leans to the left of its mean. Practically that means that its returns had some high drops in price in the sample period. Indeed, after finding the lowest value in the returns of the said company, one can discover that on the 7th of November, 2011 they had a 40.3% decrease in price. On the other hand, there is Deutsche EuroShop AG, which has a skewness of 3.25. After thorough examination, it was found that they had a sudden increase in price of 40.1% on May 23rd, 2022, which is quite recent. Albeit, the changes shown are relative, so even though they may sound colossal, they should not be such after all. Most of the sample skewness falls within the $[-1; 1]$ range, with only a few other notable exceptions.

After analysing sample excess kurtoses of the given return series, one can discover some noteworthy examples as well, for instance media company RTL Group S.A. has a sample kurtosis of 97.3, which indicates a highly leptokurtic sample distribution of price changes. Practically, it can be reasoned as an equity that has approximately the same price, without any sudden changes. Quite interestingly, previously mentioned 1&1 AG and Deutsche EuroShop AG have kurtosis of 44.7 and 72.5 respectively, which suggests that even though the stocks have sudden drops or rises in prices, they tend to be roughly stable.

The last step in the analysis of central moments should be the comparison with a normal distribution. This can be achieved with Jarque and Bera test that was described in subsection 1.2.1, where one rejects normality if the p-value is less than the significance level, which for the duration of this work will be set to $\alpha = 0.05$. After testing the given hypothesis on each stock, the following conclusion was drawn: for each stock on a given significance level, it can not be assumed that the sample returns series' distribution follows normal.

2.2 Correlation matrix analysis

The sample cross-correlation matrix of equity returns was obtained following the standard technique. The result of this is a 108×108 matrix that contains correlations between each two pairs of equities. The matrix has a unit vector on its main diagonal and is symmetrical around it. Average correlation among all pairs is 0.302, the highest average correlation belongs to the chemical producer BASF SE, which has a correlation of 0.437, on the contrary, the lowest average correlation belongs to SNP, an IT consulting and software company. Thorough examination of correlation pairs may bring following results: the highest scores belong to pairs that are from the same industry, e.g. Volkswagen AG and Porsche AG or Mercedes Benz AG and BMW AG, the lowest scores belong to the pairs that have the least in common, such as aforementioned SNP and DMG, a mining company. With this thought in mind, it was decided to analyse average correlation among the companies in the same industries. The most correlated industries are insurance, banking and auto manufacturing where the average correlation is as high as 0.891, 0.878, 0.823 respectively. However, please note that the industry groups are not represented equally in quantity. Some of the groups might be

less populous than the others. The overall number of industries accumulates to 23. All things considered, it was decided that in future applications, the sectors will be used, since they are less numerous and are distributed roughly equally.

2.3 Autocorrelation analysis

One of the widely used Portmanteau test was described in the subsection 1.2.2, the number of lags was chosen to be $\ln(3450) \approx 8$.

After using the said test, it was defined that 73 out of 107 time series exhibit autocorrelated behaviour up to the lag of 8. The presence of autocorrelation implies that the observations within each of these time series are not an IID process, since they do depend on their own past values. Therefore, for these 73 time series, one can argue that this assumption was violated.

2.4 Stationarity test

One of the most important assumption for conducting a principal component analysis on time series data is its stationarity as was described in the subsection 1.2.3, because underlying correlation or covariance matrix is reliant on the assumption that the data is stable and exhibits consistent statistical properties over time, allowing for meaningful comparisons and analyses across multiple time periods. For empirical studies, one can suffice with proving that the series is weakly stationary. The Augmented Dickey Fuller test was chosen to conduct an examination of given time series. Since the original data was preprocessed by converting the price series to price changes series, the stationarity was ensured, which was proven by the statistical test as well: all 108 variables are weakly stationary.

2.5 Principal component analysis

This section focuses on applying the principal component analysis on given data as described in the section 1.3. Both correlation and covariance matrix will be used as an underlying foundation for PCA and their results will be compared. Following topics will be subjects for comparison: explained variance, components' loadings against each other and loadings by sectors. An appropriate number of retained components will be determined by a scree plot. Lastly, both research questions will be explored further in this part.

2.5.1 Correlation matrix as an underlying foundation

The first approach takes a 108×108 sample correlation matrix $\hat{\rho}$ as an input. After following steps described in the methodology part, a 108×108 matrix was obtained, with rows indicating different stocks and columns - different eigenvectors that were sorted by their respective eigenvalues. The optimal number of principal components can be deduced from the graph 2.2.

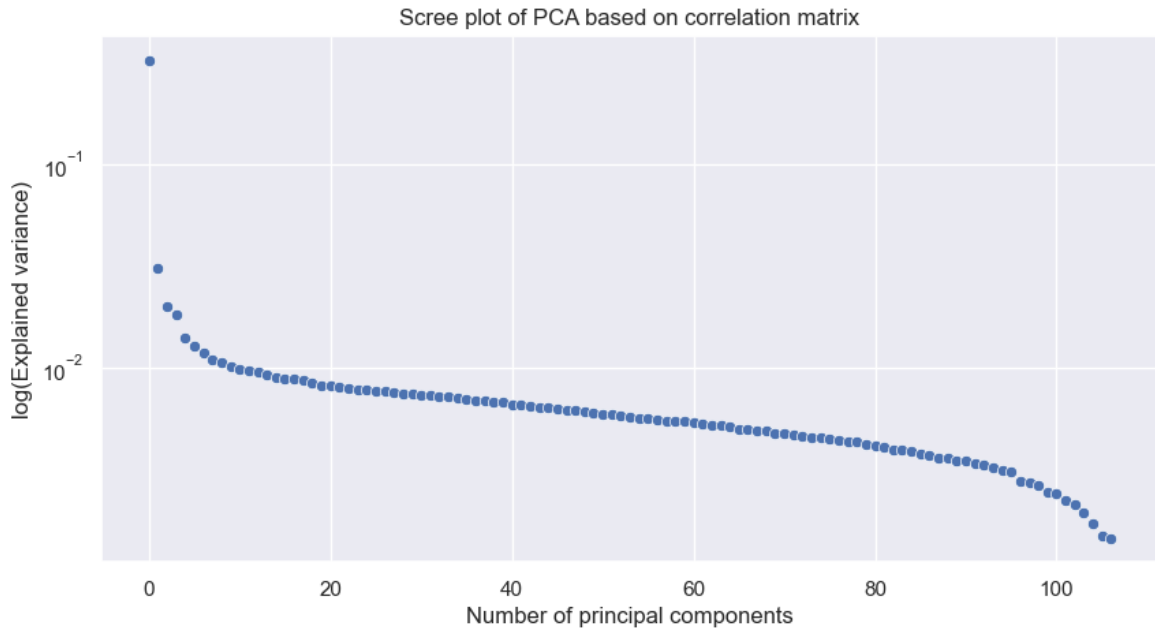


Figure 2.2: Scree plot of PCA based on sample correlation matrix $\hat{\rho}$ (Yahoo! Finance data, Author's calculation)

As can be seen on the graph 2.2, roughly around the tenth principal component, the explained variance starts to be steeper, so that adding extra principal component will not be as beneficial as it could have been. This number also coincides with the number of industry sectors present in the data, as was suggested in the section 1.4. With that in mind, if an investor decides to expose themselves to almost half of the risk on the given market, they should allocate their risk budget according to the eigenvectors of the first 10 principal components.

The table 2.1 suggests that the first component explains the most of the variance singlehandedly: 32.5% of whole variance in the sample. All additional components explain less than 4% each. This is not an usual behaviour for principal component analysis on panel data, where one would not expect such a sudden drop in explainability, albeit for time series data, this might make sense, since co-movement is tried to be explained. The obtained results are in line with the findings of other authors that have conducted their research on the same topic, most notably (Avellaneda and Lee, 2010), where the authors have explored the U.S. equity market and have obtained the first component that had explained 20% of overall variance with all succeeding components explaining less than 4% each. The first component has an interest interpretation, (Jolliffe, 2002) calls it a “size” component, whilst (Tsay, 2010) gives it

Table 2.1: Cumulative explained variance of first 10 components of the PCA based on sample correlation matrix $\hat{\rho}$

Principal component	Cumulative explained variance
1	0.325
2	0.356
3	0.376
4	0.394
5	0.408
6	0.421
7	0.433
8	0.444
9	0.454
10	0.465

a “market” component name, since it represents the general movement of the equity market. Most notably, (Laloux et al., 2000) state that the first principal component can be considered to be a "market" component, since it has roughly equal elements on all the N stocks in its eigenvector. Another way to examine the principal components further is to look at their distributions and interactions among each other.

On the graph 2.3, on the main diagonal the distributions of each principal component’s eigenvectors are displayed. The off-diagonal entries contain scatter plots of the eigenvectors’ values against each other. The colour indicates volatility of a given equity: the darker it is, the more volatile is the financial asset.

One can notice that the histogram of the first principal component’s eigenvector resembles a symmetric distribution around its mean. Quite surprisingly, the scatter plot of the first principal eigenvector against the second one hints that there is a negative correlation between them. The interactions between the first and the third and the fourth one do not seem to indicate any relationship between them whatsoever. The fifth principal eigenvector on the other hand does resemble a positive relationship with the first one. Volatility does not seem to play a big role in the loadings of an eigenvector. The second principal eigenvector’s histogram suggests that the distribution is lightly skewed to the right, towards the negative loadings. Other than with the first principal eigenvector, the second one does not seem to interact with other eigenvectors, since the scatter plots do not indicate any relationship. The distribution of the third principal component’s eigenvector can be proven to be quite peculiar: it is skewed to the left, towards more positive loadings, albeit there are still some negative values present. The interactions with other eigenvectors does not seem to be of any significance either. From the histogram of the fourth principal component’s eigenvector, it is evident that the preference is being given to small positive values, as indicated by distribution’s mode. One can argue that the histogram of the fifth principal eigenvector has a resemblance of the

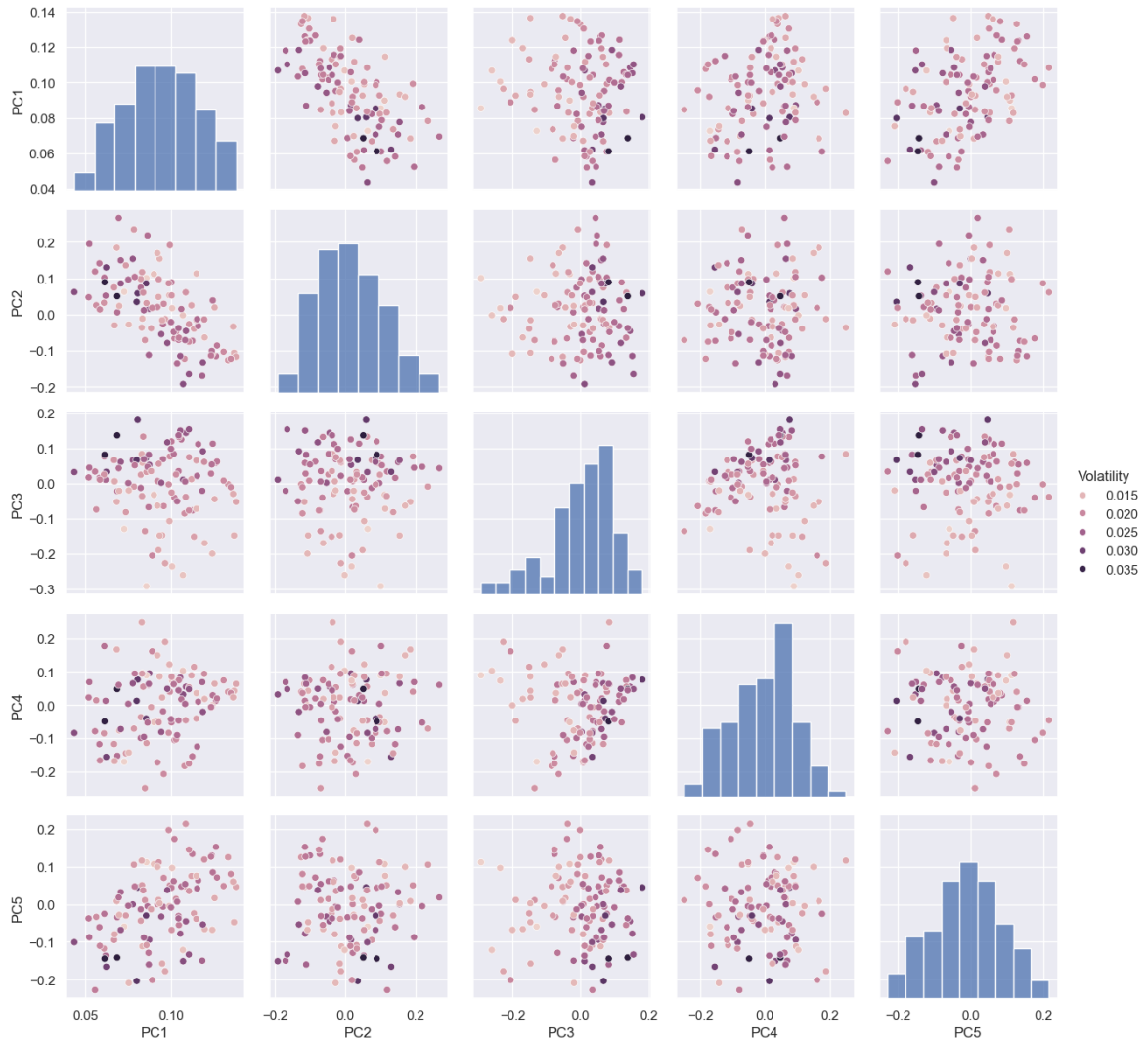


Figure 2.3: Interactions between the first five principal components based on sample correlation matrix $\hat{\rho}$ (Yahoo! Finance data, Author's calculation)

normal distribution with mean 0.

Table 2.2: Correlation coefficients between the first five principal eigenvectors based on sample correlation matrix $\hat{\rho}$

	e_1	e_2	e_3	e_4	e_5
e_1	1.000	-0.623	-0.102	0.209	0.438
e_2		1.000	-0.003	0.007	0.014
e_3			1.000	0.001	0.002
e_4				1.000	-0.005
e_5					1.000

The table 2.2 supports the aforementioned ideas: some of the eigenvectors are indeed correlated with the first principal eigenvector. As was stated previously, the first principal component may indicate overall market movement, since it explains the most variability of the data, so that means that some of the other eigenvectors are still exposed to overall market trends, which would make sense. On the contrary, the other four eigenvectors are not anyhow correlated among each other.



Figure 2.4: Eigenvectors of the first five principal components based on sample correlation matrix $\hat{\rho}$, coloured by their respective sectors (Yahoo! Finance data, Author's calculation)

The eigenvectors' values of the first five principal components are visualised on the graph 2.4.

Immediately one can notice that the first component distributes the loadings roughly equally among all stocks, this almost coincides with the portfolio with equal weights ($1/N$), and even further supports the idea that the first principal component is indeed a market component indicating overall movement, as was stated in the work of (Laloux et al., 2000). As one can see from the colours, no particular sector was favoured over the other and all the loadings are positive, which indicates that the investor is supposed to go long on them. The second principal component has fundamentally different eigenvector structure than the first one. It is noticeable that the loadings are not even roughly equal, and some negative scores are present as well as opposed to positive-only first component. This component gives positive scores to most of the stocks coming from Healthcare and Technology sector, and negative ones to equities from Basic Materials, Consumer-related or Industrials. Following (Tsay, 2010) classification, one may call this component a "Healthcare + Technology industry vs. Basic Materials + Industrials", which basically means that one should go long on the first pair and go short on the second one, as defined by the respective weights. The third component may be similarly described as an industry component, however this time it favours Industrials and Technology sectors for long positions, and Communication, Consumer Defensive and some of the Financial companies for short positions. As opposed to the second principal component, that gave big weights for long positions, the third one gave bigger weights for multiple short positions, and moderate weights for the long ones. The fourth and fifth principal components clearly do not favour any particular sector or industry over the others, since they may go both long and short on the same sector, in seemingly random manner. In this case it would be beneficial to look at another company characteristics, such as their market capitalisation or company size, or to go deeper into classification, thus dividing the stocks by their immediate industry, to determine what logic may stand behind those principal portfolios construction.

2.5.2 Covariance matrix as an underlying foundation

The second approach follows the same principles as the first one, only that this time, the sample covariance matrix $\hat{\Sigma}$ is to be used as an underlying foundation for PCA, following the process described in the section 1.3. A given output is a 108×108 matrix, with rows being different stocks from the original dataset and columns - eigenvectors that were sorted by their respective eigenvalues.

The optimal number of principal components can be deducted in a similar manner as previously, using a scree plot on the graph 2.5. The explained variance decreases steeply at first, one can notice a sudden drop in explanatory power between the first and the second principal components, however every next component adds less and less explanation of the variance, which results in the curve flattening. Although it is not quite obvious from this graph, but an investor may decide to choose first 10 principal portfolios in a similar manner as with the correlation matrix approach.

As can be seen on the table 2.3, the first principal component is successful in explaining one third of the overall variance present in the data. The situation with the succeeding principal

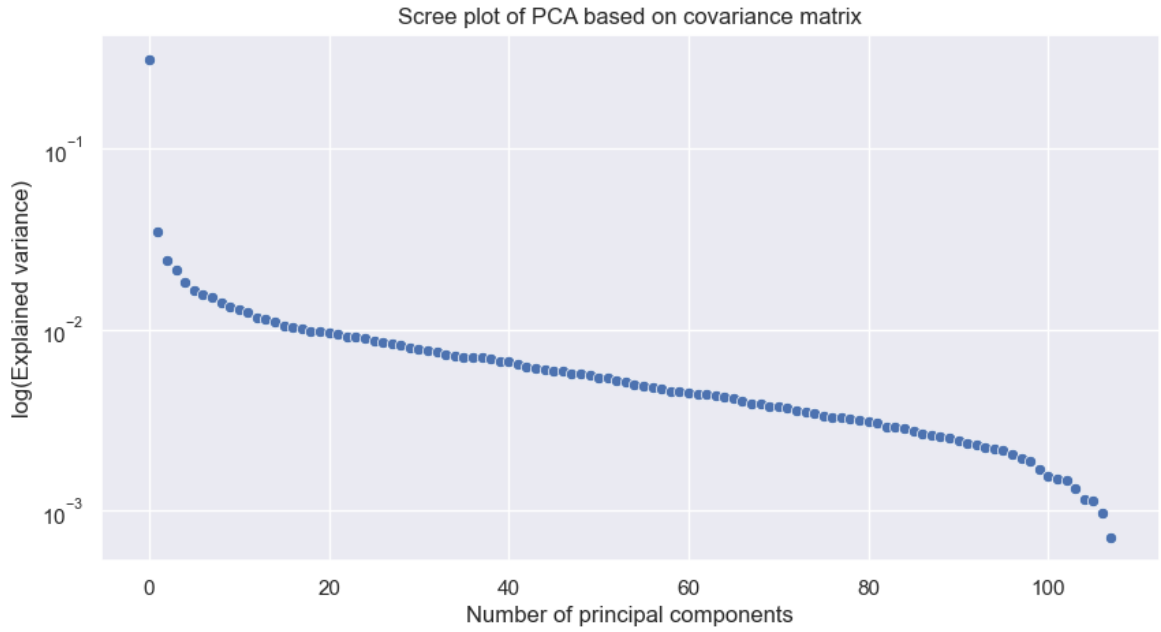


Figure 2.5: Scree plot of PCA based on sample covariance matrix $\hat{\Sigma}$ (Yahoo! Finance data, Author's calculation)

Table 2.3: Cumulative explained variance of first 10 components of the PCA based on covariance matrix $\hat{\Sigma}$

Principal component	Cumulative explained variance
1	0.312
2	0.346
3	0.371
4	0.392
5	0.410
6	0.427
7	0.442
8	0.457
9	0.471
10	0.485

components and their explanatory power is similar to the correlation matrix approach: each additional component adds less than 4% to the explained variance. Clearly, both approaches are roughly identical on this matter.



Figure 2.6: Interactions between the first five principal components based on covariance matrix (Yahoo! Finance data, Author's calculation)

The subsequent items to be discussed are the eigenvectors, their distributions and interactions with each other. The aforementioned was depicted on the graph 2.6. The first principal component's eigenvector exhibits roughly the same behaviour that was observed in the case of the correlation matrix - the loadings are all positive ranging from 0.05 to 0.15 with no clear mode. The distribution of the first component's eigenvector against the second one's is quite peculiar, since one can observe a positive linear relationship between these two variables. Also, volatility seems to play a role here as well, a gradient pattern is noticeable here, with lower volatility stocks gathered in the lower right corner of the plot, and the higher volatility equities - in the higher left. Which basically means that, whilst the first principal eigenvector gives higher positive loadings to higher volatility equities, the second principal eigenvector assigns them with high negative scores. Practically, the second principal component's eigenvector can be said to go short on high volatility stocks, and go long on middle volatility

equities, as can be seen on the graph. Examination of the distribution of the second principal component's eigenvector can entertain a theory that most of the stocks are being ignored, which is indicated by its mode directly at 0. On the first glance, it is noteworthy that the third principal eigenvector has a few outliers: one can notice that there are 3 equities that are assigned significantly higher scores than the rest of the asset universe, one of them is as big as 0.75, which skews the distribution towards the left, with a sample skewness coefficient of -0.855 . The distribution of this eigenvector has a well defined peak around 0, which is further supported by a positive kurtosis of 0.591, and only a small number of equities has negative scores. Practically, if an investor decides to allocate their assets according to this principal eigenvector, they will have to put a substantial portion of one equity on a long position, then some relatively smaller parts on the other two outliers, and roughly ignore the rest, with no short positions. Even though the fourth principal component's eigenvector has outliers as well, they are not distributed in a similar fashion to the third one. This time, the component suggests to go short on one volatile asset, which score exceeds 0.8, and compensate that with two long positions on another highly volatile equities. This eigenvector also overlooks less risky stocks, assigning them values that are not any different from 0. The distribution of the loadings for this eigenvector also exhibits a clear peak at 0. One can argue that the fifth principal eigenvector is not as extreme as the previous two, albeit it still favours riskier assets over the less volatile ones. It assigns a score of 0.5 to one volatile asset, practically suggesting to go long on it, and a score of -0.25 on another volatile asset, going short. One interaction between two principal eigenvectors is distinguishably peculiar: if an investor decides to allocate their risk budgets following the third and the fourth principal components, they will be exposed to almost all risky equities in the asset universe, practically ignoring the rest of it.

Table 2.4: Correlation coefficients between the first five principal components's eigenvectors based on sample covariance matrix $\hat{\Sigma}$

	e_1	e_2	e_3	e_4	e_5
e_1	1.000	0.416	0.401	-0.040	0.190
e_2		1.000	-0.015	0.001	-0.007
e_3			1.000	0.001	-0.007
e_4				1.000	0.001
e_5					1.000

The correlation matrix presented in table 2.4 suggests that, similarly as in the previous case, principal components' eigenvectors based on covariance matrix display an independent nature, beside the very first eigenvector. It has a slight positive relationship with the second and the third principal eigenvectors, all in magnitude of 0.4. However, all other eigenvectors are indeed not intercorrelated.

The loadings of each principal component's eigenvector are visualised on 2.7. As one can notice, the loadings on the first component's eigenvector are distributed in a similar fashion as previously, resembling a portfolio that could track the market, since all loadings are roughly same in size, despite the assigned sector. The second principal component's eigenvector

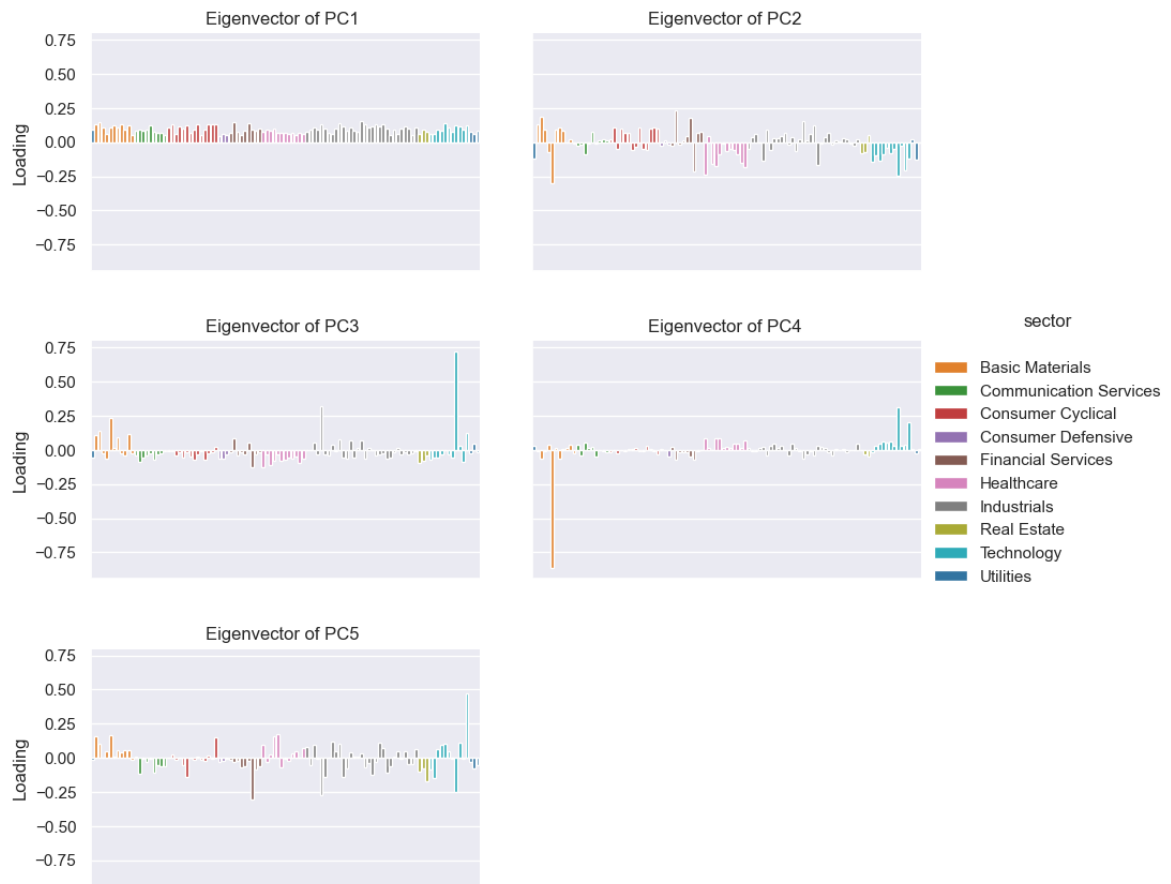


Figure 2.7: Eigenvectors of the first five principal components based on sample covariance matrix $\hat{\Sigma}$, coloured by their respective sectors (Yahoo! Finance data, Author's calculation)

though looks nothing alike as its counterpart from the previous section: firstly, it does not assign loadings to all of the available stocks, giving favour to only a handful of them. Although it is not quite clear, nevertheless one can notice that this eigenvector suggests to go short on healthcare and technological sector, in a similar manner as the one in the previous section. The third component's eigenvector, as was mentioned before, has extreme values, it clearly goes long on a company coming from the technological sector, and then on the one in the industrial sector and a handful of companies from the "Basic Materials". Clearly, it almost does not go short, except for some smaller scale values in the healthcare sector. The fourth principal component's eigenvector, on the contrary, favours one short position on a company from "Basic Materials", which it assigns quite a high negative weight, then it goes long on the equities from the technological sector. As was mentioned before, the fifth principal component's eigenvector is rather mild in assigning the loadings, compared to the previous two. Nevertheless, it still does assign a fairly high positive loading to one equity from the technological sector. Other than that, it is not evident if a pattern of assigning those loadings can be established, since it can go both long and short on the same sector.

2.5.3 Assessing the principal portfolios

After examining the eigenvector elements' distribution, one can go further to finally assess the principal portfolios themselves. Using formulae 1.11 and 1.12, the principal portfolios were constructed, effectively producing 5 distinct portfolios for each approach. The comparison was done following the methodology of (Lohre et al., 2012), where the authors were using cumulative returns. Comparison of the first principal portfolios with the benchmark is displayed on the graph 2.8. The DAX Performance Index was chosen as the benchmark for the German equity market. It is evident from the figure, that both portfolios and the benchmark do have similar curves, albeit the benchmark falls short. One may conclude that the first principal components based on both approaches are more performant than the official benchmark. Both first portfolios are quite similar in their dynamics, albeit the one based on covariance matrix has shown better performance since the mid-2020. All this further proves that despite the approach, the most dominant eigenvalue produces the eigenvector that will mimic the market, hence the name "market" component.

The cumulative returns' development over time is displayed on the graph 2.9. The X-axis reflects the date and the Y-axis corresponds to cumulative returns, different portfolios are colour-coded as shown on the graph's legend. As one can notice from the very first glance, from the first five principal portfolios only the fourth one is unprofitable, as its cumulative returns fall below 0 to the end date. All others do generate different magnitudes of profit. Quite interestingly, the second principal portfolio was underperforming from 2010 to 2020 compared to the first one, then the second component exceeded its counterpart briefly for less than a year, and then fell below again. This sudden spike in performance of the second principal component and decline of the first one might find an explanation in the world-wide COVID epidemic, since the second principal component has a significant number of its weights allocated to long positions on Healthcare and Technology sector, whilst the first one

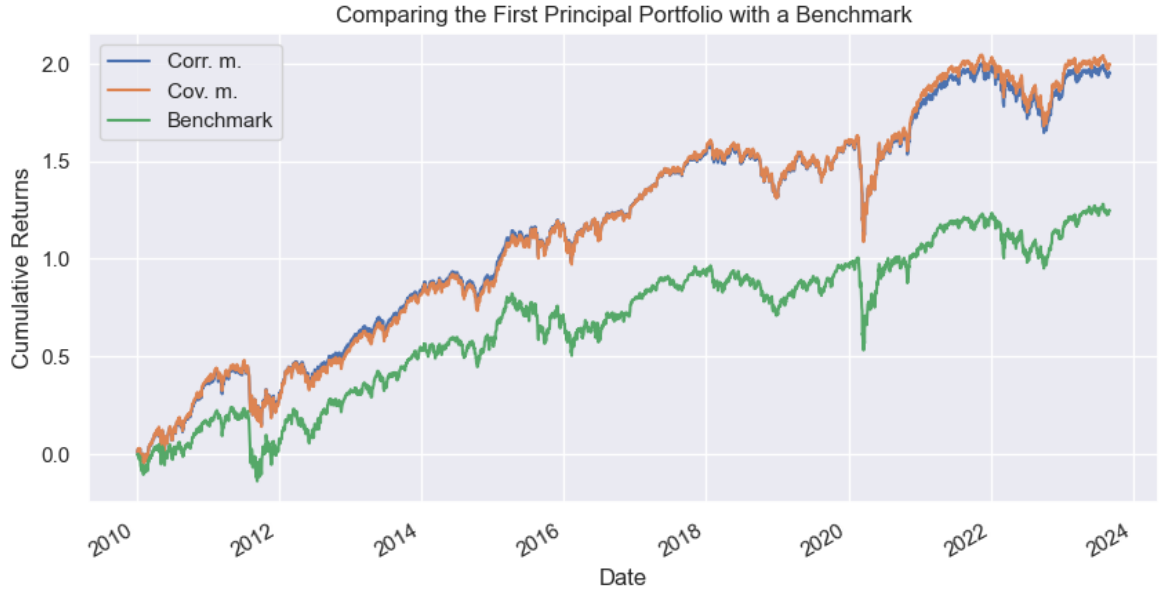


Figure 2.8: Cumulative returns of the first principal portfolios based on both approaches compared with the DAX benchmark (Yahoo! Finance data, Author's calculation)

distributes its weights roughly equally, thus reflecting the overall situation of the market. The third and the fifth principal components are less performant than the first two, overall they end up having around 0.5 cumulative return to date.

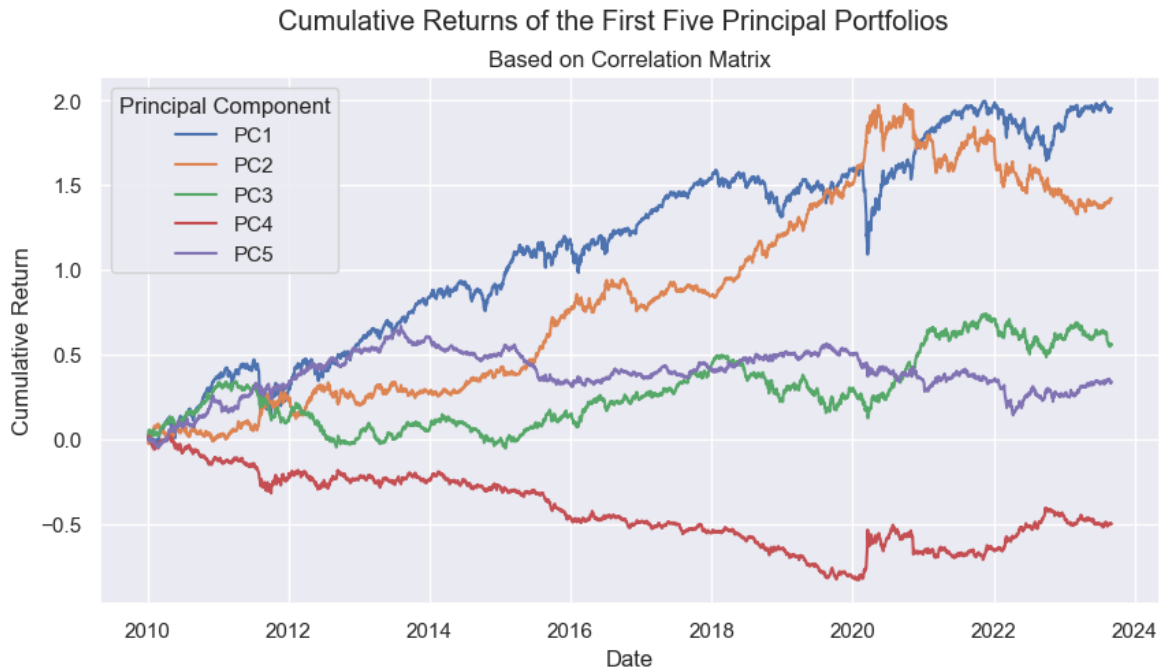


Figure 2.9: Cumulative returns of the first five principal portfolios based on sample correlation matrix $\hat{\rho}$ (Yahoo! Finance data, Author's calculation)

The graph 2.10 displays cumulative returns of the first five principal components based on sample covariance matrix $\hat{\Sigma}$. The structure of the figure is similar to that of 2.9. This time,

only the very first principal component is profitable, as it reaches a cumulative return of roughly 2.0 to date. The fifth principal component has positive returns as well, albeit they fall beneath 0.5. All other three components are negative in their returns, with the second principal component being the lowest in cumulative returns. The other two portfolios fall between $[-1, 0]$ range. With all that being said, one may conclude that the first five principal components based on covariance matrix are less profitable as compared to their counterparts based on sample correlation matrix.

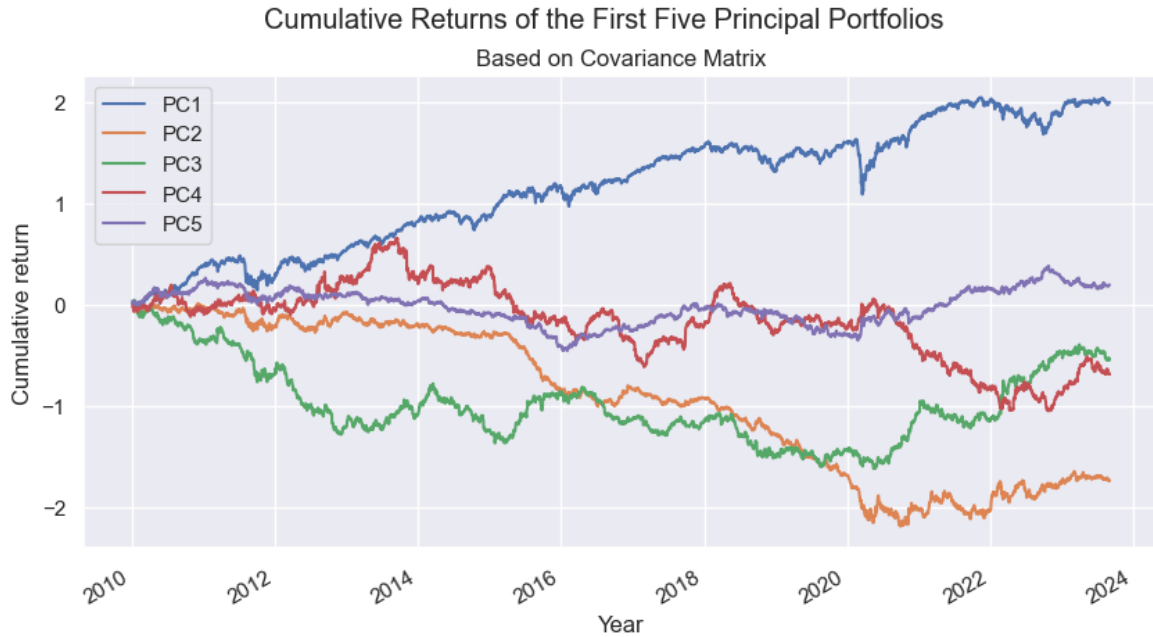


Figure 2.10: Cumulative returns of the first five principal portfolios based on sample correlation matrix $\hat{\rho}$ (Yahoo! Finance data, Author's calculation)

2.5.4 Interpretations of Principal Portfolios

The second research question that was proposed in this thesis was to find a possible interpretation for the first N components. There are multiple ways how to do that, one of which was explored in the sections 2.5.1 and 2.5.2, where it was established that some of the principal portfolios go long on some industries and go short on another ones, effectively maintaining a pattern of being industry components. However, as it was previously mentioned, this methodology may have some drawbacks, since it considers only one dimension of exploring the components, which is by their industry sector affiliation. Another approach was explored in Methodology in section 1.5, which requires conducting a linear regression analysis on principal portfolios against already established and known factors. Using an equation 1.14, one may construct a linear regression model. As was mentioned in the theoretical discussion, there is no need to construct the Fama-French factors from scratch, since the authors kindly provide the return series' matrix $\mathbf{R}_{FF}$ on their website. The only required adjustment is to match the time window to the one that was used for constructing the principal portfolios. After following the described steps, one may acquire the results in table 2.5

Table 2.5: Linear regression analysis results of the model 1.14 for the principal portfolios for the period from the beginning of 2010 till September, 2023.

	PC_1	PC_2	PC_3	PC_4	PC_5
<i>Coefficients</i>					
Intercept	0.000	0.000	0.000	0.000	0.000
Mkt-RF	0.857	-0.157	0.351	-0.027	0.042
SMB	-0.052	0.294	0.836	-0.562	-0.303
HML	0.191	-1.292	0.197	-0.049	0.119
RMW	0.165	-0.216	-0.136	-0.381	0.498
CMA	-0.210	-0.080	-0.506	-0.208	-0.283
<i>p-values</i>					
Intercept	0.002	0.000	0.508	0.457	0.944
Mkt-RF	0.000	0.000	0.000	0.027	0.000
SMB	0.073	0.000	0.000	0.000	0.000
HML	0.000	0.000	0.000	0.141	0.000
RMW	0.001	0.000	0.011	0.000	0.000
CMA	0.000	0.118	0.000	0.000	0.000
<i>Adjusted R^2</i>	0.742	0.602	0.292	0.146	0.116

The table 2.5 displays the results of the linear regression analysis conducted on principal portfolios created by sample correlation matrix approach. As one can notice, for the very first principal component the explained variance accumulates to 74.2%. With $Mkt - RF$ having the biggest coefficient out of all other variables, this may hint that for the first principal component, the main driver of the returns can be stated a *Market Risk Factor*, which even further proves the aforementioned theory, that the first principal component is a "market" component, effectively confirming the idea. The said fact implies that the first principal component represents the biggest portion of systematic risk that is present on the German equity market. The other Fama-French factors do not have such a big impact. All the factors are statistically significant, except the *SMB* one, which is proven by their respective p-values. The second principal component has an adjusted R^2 of 0.602, this time one can notice a drop in explainability, albeit not that immense, which suggests that there might be some other factors in play that were not considered in this analysis. This time the highest absolute value belongs to the *HML* factor, but with a negative sign. This suggests that these two variables might have an inverse relationship, effectively stating that one may consider constructing a *Low minus High* portfolio to achieve a positive sign in this case. The second most contributing factor is *SMB* with a value of 0.294, which this time has a positive linear relationship with the dependent variable. Every single variable is statistically significant, except the *CMA*. The third principal component has an adjusted R^2 of only 0.292, which may indicate that the given factors are not sufficient for explaining the variance of the returns of a given principal

portfolio. Nevertheless, it is noteworthy that the *SMB* coefficient has the highest value of 0.836, with $Mkt - RF$ having the second highest coefficient of 0.351. These two variables may be said to be the main drivers of the explained portion of the variance. All of the variables are statistically significant, except for the intercept. Considering this, the intercept does not prove to be significant in any further principal components as well. The fourth principal component's adjusted R^2 equals to 0.146. Quite interestingly, all of the coefficients, despite the intercept, are negative in their sign. *SMB* and *RMW* have the highest loadings in absolute value of -0.562 and -0.381 respectively. As was previously mentioned, this might have an explanation in a way the factors were constructed: if the sign were to be reversed, e.g. *RMW* would become *Weak minus Robust* etc., then the positive signs might have been achieved. Thus one may state, that the explanation of the fourth component might be that it is *Big minus Small* and *Weak minus Robust* with adjusted proportions. From all five factors only the *HML* is not significant. The fifth principal component has the lowest adjusted R^2 out of all aforementioned portfolios - only 0.116. These regression coefficients are all statistically significant, despite the intercept, as indicated by their respective *p-values*. This time the coefficients do not have a uniform sign, as two of them are negative, and the three others are positive. The highest absolute value belongs to *RMW* - 0.498, the second highest is again *SMB* with the value of -0.303 . The fifth principal portfolio might be explained as a positive *RMW* versus negative *SMB*, albeit both these coefficient explain only 11% of overall variance. The obtained results are in line with those of (Lohre et al., 2012), where the authors achieved the same outcomes, they proposed that most of the principal portfolios' time series variation cannot be accounted for by the common factors only, since either the higher order principal components are not meaningful or the factor structure of 1.14 is incomplete and may be lacking some important factors.

2.5.5 Comparing both approaches

The first research question of this thesis is to deduct what is the difference between using correlation and covariance matrix as an underlying foundation for the PCA in the context of financial markets. This subsection will further explore both approaches and compare them. The very first thing that can be evaluated is the explanatory power of both of them, for this purpose, the difference between explained variance for the first ten principal components is found.

The table 2.6 was constructed using formula $PC_{cov} - PC_{corr}$, i.e. the explanatory power of the principal components of covariance matrix minus those of correlation matrix. As can be clearly seen from the table, the very first principal component explains more variance in case of correlation matrix, albeit only by a fraction. All succeeding principal components of the covariance matrix carry more explanatory power than their correlation counterparts, even though each of them differs only by roughly 0.004, this explanatory power debt builds up, as one goes to higher-order principal components. Undoubtedly, the covariance matrix serves as a more efficient fundamental basis when aiming to explain a significant portion of variance while minimising the number of principal components is the goal. While examining

Table 2.6: Differences between the explanatory power of the first 10 components from both approaches

Principal component	Difference in explained variance
1	-0.013
2	0.004
3	0.004
4	0.003
5	0.004
6	0.004
7	0.004
8	0.004
9	0.004
10	0.003

the eigenvectors' values in the two subsections 2.5.1 and 2.5.2, one could not help but notice that the covariance matrix approach clearly favours the equities with higher volatility. This results in principal components that have high loadings for a handful of equities that are volatile and complete disregard for other assets. On the contrary, the correlation matrix approach does a better job in spreading the loadings roughly evenly across all given asset universe, thus eliminating any outliers.

Table 2.7: Comparison of diversification ratio $DR(w)$ of the long positions of the first five principal portfolios for both approaches

Principal portfolio	Diversification ratio $DR(w)$ for	
	correlation matrix	covariance matrix
1	1.725	1.745
2	1.928	1.431
3	1.761	1.571
4	1.735	1.771
5	1.540	1.711

The table 2.7 displays the diversification ratios $DR(w)$ based on formula 1.13 for both approaches of constructing the principal portfolios. As can be seen, the first principal portfolios differ only by 0.02, with the correlation matrix approach being less diversified than its counterpart. The second principal portfolio based on correlation is clearly more diversified than its counterpart as it has a higher diversification ratio by roughly 0.5. The third principal portfolio is more diversified in case of correlation matrix approach once again. The fourth and the fifth principal portfolio are more diversified, when the covariance matrix is used as an underlying foundation. Clearly, for the first five principal components there is no clear winner, albeit on average the correlation matrix approach produces more diversified portfolios.

In the subsection 2.5.3 the cumulative returns of the first five principal portfolios were compared. As it was stated there, the correlation matrix approach subsequently produces portfolios that are more profitable than their covariance matrix counterparts.

Conclusion

In the intricate landscape of capital markets, the interplay between risk and their returns is a fundamental consideration for investors and financial analysts alike. Multiple practitioners delve into the topic of financial risk, researching different aspects of it. The employment of principal component analysis on financial data has a wide use, since it allows to deconstruct the risks into a smaller subset of the original asset universe. At the outset of this study, the central goals that guided the investigation were to reveal if the standardisation of the returns has any effect on constructing the eigenportfolios, and if the constructed principal portfolios might have any interpretations, thus uncovering the underlying risk sources.

Based on the conducted exploratory analysis in the section 2.1, the mean - volatility structure of the given asset universe was uncovered, stating that a big portion of the equities has mean return less than 0.008 and volatility below 0.025. The returns' distributions of the assets were examined further to reveal that their kurtosis and skewness has peculiar values that indicate how the assets behave in the given timeframe. Lastly, the said distributions were tested against the normal distribution, where it was proven that none of the assets' returns indicate normality. The sample cross-correlation analysis of the returns was conducted in the section 2.2, where it was revealed that the average correlation coefficient between the assets' returns is 0.302. It was also uncovered that the companies from the same industry have the highest correlation among them, e.g. Volkswagen AG and Porsche AG. With this thought in mind, it was also decided to examine the cross-correlation between the companies coming from the same sector, which led to following insights: the most intercorrelated industries are insurance, banking and automobile manufacturing, where the average correlation coefficients rise as high as 0.891, 0.878, 0.823 respectively. The stationarity test in the section 2.4 revealed that the most important assumption for conducting the principal component analysis, having the underlying data stationary, was successfully fulfilled, thus ensuring that all following analysis is correctly specified.

The section 2.5 served as an umbrella for broader subtopics of principal component analysis that were imposed in this thesis. The subsections 2.5.1 and 2.5.2 provided the analyses for the eigenvectors based on correlation and covariance matrix respectively. There it was discovered that the first 10 components explain roughly half of the variance of the original asset set for both approaches. The very first component carries around 1/3 of the original variance, with all subsequent components bringing less than 4% to overall explainability. The distributions of the eigenvectors were examined further, both approaches leveraged the possibility to create both positive and negative values, thus indicating long and short weights respectively. The first principal component has a positive-only eigenvector that has roughly equal loadings, multiple sources arrived on the same result, deriving the conclusion that the said component can be labelled as a "market" component, thus implying the the first principal component displays a risk source coming from the market movement itself. Furthermore, the correlation between the first five principal components' eigenvectors was analysed, both

approaches had intercorrelated eigenvectors, albeit only with the first principal component, indicating that some of the portfolios might still have a systematic risk present. The correlation and covariance matrix approach diverged in the eigenvectors' loadings allocation: the standardisation of the returns allowed for more equal weights distribution, whereas the lack of thereof enabled the high volatility assets to gain bigger weights, thus creating dominant equities in each portfolio. Later, the eigenvectors' values were visualised by their sectors, to see if the portfolios are somehow dependent on the industry a certain equity is coming from. As was stated previously, the first eigenportfolio has roughly equal positive weights for all stocks. It was revealed that the second principal portfolio can be interpreted as "Healthcare and Technology" for long positions vs. "Basic Materials and Industrials" for short positions in case of correlation matrix. The third principal component favours Industrials and Technology for long positions, and Communication and Consumer Defensive for short. The fourth and fifth principal components can not be said to clearly favour one sector over the other, thus it was proposed that the underlying variance may come from other sources, not only from the respective sectors. As was previously mentioned, the covariance matrix approach has some equities that dominate over the rest of the asset universe, which may obstruct in defining a clear picture in case of which sector is long, and which is short. Albeit, the second principal component goes short on Healthcare and Technology. The third principal component goes long on one company from the Technology sector and one from Industrials sector, and then goes short on Healthcare sector. The fourth principal component has a dominant short position on a company from Basic Materials, and then it goes long on Healthcare and Technology. The fifth principal component does not seem to have a clear defined structure based on sectors, as was the case previously.

The subsection 2.5.3 assessed the principal portfolios that were constructed based on the weights given by both approaches. Here it was revealed that both covariance and correlation matrix approaches produce the first component that has better cumulative return than the German market benchmark. Both first components are very similar in their behaviour, albeit the covariance matrix approach produces better returns since the middle of 2020, as was examined during the backtesting. However, the eigenportfolios produced by correlation matrix have better cumulative returns compared to their covariance matrix counterparts, with the second principal portfolio having better returns for a short period of time.

Lastly, in the subsection 2.5.4 the principal portfolios produced by a correlation matrix were compared against the predefined Fama-French 5 factors model as was defined in the section 1.5. Here it was confirmed that the first component is indeed a "market" component, since the respective linear regression coefficient has a value of 0.857, all other coefficients do not exceed a magnitude of 0.2 in absolute value. The second principal component had a negative loading on *HML* factor with a value of -1.292 , effectively meaning that this principal component may be labelled as *Low minus High*. The third principal component had the highest coefficient on the *SMB* factor with a value of 0.836, thus getting an interpretation of the *Small minus Big* component. The fourth and the fifth principal component had their biggest absolute coefficients on *SMB* and *RMW* coefficient respectively. With the fourth principal component getting a *Big minus Small* label, and the fifth *Robust minus Weak* label. Albeit, the overall

variance of these components was explained only partly - roughly 10% in both cases, as was indicated by their respective R^2 , the possible implications were explored further in the aforementioned subsection.

After comparing both approaches, one can be ready to answer the first research question: indeed, when speaking of financial time series, it is important to standardise the returns beforehand, in order to produce the portfolios that have their weights allocated evenly. This positively contributes to the diversification ratio as well, as was compared in the subsection 2.5.5. Moreover, the correlation matrix approach produced portfolios that performed better during the backtesting.

As was explored throughout the duration of the thesis, the first principal component can be interpreted as a market component, despite the chosen approach. For given timeframe and asset universe, the second principal component can be effectively interpreted as a "Healthcare and Technology" vs. "Basic Materials and Industrials" and also a *Low minus High* component, thus identifying and leveraging the said risk sources. The third component can be labelled as "Industrials and Technology" vs. "Communication and Consumer Defensive" and *Small minus Big* component. The fourth principal component could not be linked with some particular sectors, albeit it can still get a *Big minus Small* label. The fifth principal component did not have any affiliations with sectors structure as well, however it can still be interpreted as *Robust minus Weak* component. Effectively, these are the top five dominant risk sources for the German equity market for a given timeframe.

Nevertheless, the said thesis had some limitations that could be addressed in future research. Firstly, the whole time window was considered during the portfolio construction, whereas multiple practitioners conduct the said methodology by picking a rolling window of a market cycle, and reconstructing the portfolios accordingly. The said limitation arose from a lack of index structure for the last 10 years, i.e. only the latest constituents are publicly available, this also implies a forward-looking bias. Secondly, the Fama-French factors that were defined in the section 1.5 and obtained from one of the authors' website are constructed for almost whole European market, even though one may argue that the EU equity markets are quite interconnected, it could still benefit the research to have a German market-only factors. The lack of publicly available respective data for the factors construction is the main obstacle in this case. If one is to conduct a similar research, building up on this work, it would be highly beneficial to examine whole EU market, since it may be proven to be quite a peculiar study. Also, the comparison of factor investing against principal portfolios may constitute an interesting future research direction, as was briefly mentioned in the section 1.4. By addressing the aforementioned limitations, one can derive different results, and conduct a completely distinct work that may serve as a foundation for another research topic.

All things considered, the goals of this thesis were achieved. The research underscores the importance of standardising the asset returns before conducting a principal component analysis on financial time series data by making a comparison between both approaches. This study also contributes insights to revealing the underlying risk sources that were present on

the German equity market for the period of the last 10 years. Furthermore, it is important to explore the ramifications of the given methodology for the portfolio construction field in the future, since principal component analysis can be proven to be a valuable tool for constructing the portfolios and identifying the risk sources.

Bibliography

- AVELLANEDA, Marco, HEALY, Brian, PAPANICOLAOU, Andrew and PAPANICOLAOU, George, 2020. Principal Eigenportfolios for U.S. Equities. *SSRN Electronic Journal*. ISSN 1556-5068. Available from DOI: <https://dx.doi.org/10.2139/ssrn.3738769>.
- AVELLANEDA, Marco and LEE, Jeong-Hyun, 2010. Statistical Arbitrage in the U.S. Equities Market. *Quantitative Finance*. Vol. 10, no. 7, pp. 761–782. ISSN 1469-7696. Available from DOI: <https://doi.org/10.1080/14697680903124632>.
- BOX, G. E. P. and PIERCE, David A., 1970. Distribution of Residual Autocorrelations in Autoregressive-Integrated Moving Average Time Series Models. *Journal of the American Statistical Association*. Vol. 65, no. 332, pp. 1509–1526. ISSN 1537-274X. Available from DOI: <https://doi.org/10.2307/2284333>.
- CIPRA, Tomáš, 2015. *Praktický průvodce finanční a pojistnou matematikou*. 3rd ed. Ekopress. ISBN 978-80-87865-18-7.
- CONNOR, Gregory, 1995. The Three Types of Factor Models: A Comparison of Their Explanatory Power. *Financial Analysts Journal*. Vol. 51, no. 3, pp. 42–46. ISSN 1938-3312. Available from DOI: <https://doi.org/10.2469/faj.v51.n3.1904>.
- DICKEY, David A. and FULLER, Wayne A., 1979. Distribution of the Estimators for Autoregressive Time Series With a Unit Root. *Journal of the American Statistical Association*. Vol. 74, no. 366, pp. 427–431. ISSN 0162-1459. Available from DOI: <https://doi.org/10.2307/2286348>.
- FAMA, Eugene F. and FRENCH, Kenneth R., 1993. Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics*. Vol. 33, no. 1, pp. 3–56. ISSN 0304-405X. Available from DOI: [https://doi.org/10.1016/0304-405X\(93\)90023-5](https://doi.org/10.1016/0304-405X(93)90023-5).
- FAMA, Eugene F. and FRENCH, Kenneth R., 2015. A Five-Factor Asset Pricing Model. *Journal of Financial Economics*. Vol. 116, no. 1, pp. 1–22. ISSN 0304-405X. Available from DOI: <https://doi.org/10.1016/j.jfineco.2014.10.010>.
- CHOUEIFATY, Yves and COIGNARD, Yves, 2008. Toward Maximum Diversification. *The Journal of Portfolio Management*. Vol. 35, no. 1, pp. 40–51. ISSN 1556-5068. Available from DOI: <https://dx.doi.org/10.2139/ssrn.4063676>.
- JARQUE, Carlos M. and BERA, Anil K., 1987. A Test for Normality of Observations and Regression Residuals. *International Statistical Review / Revue Internationale de Statistique*. Vol. 55, no. 2, pp. 163–172. ISSN 0306-7734. Available from DOI: <https://doi.org/10.2307/1403192>.
- JOLLIFFE, I.T., 2002. *Principal Component Analysis*. 2nd ed. Springer. Springer Series in Statistics. ISBN 978-0-387-95442-4.

- LALOUX, Laurent, CIZEAU, Pierre, POTTERS, Marc and BOUCHAUD, Jean-Philippe, 2000. Random Matrix Theory and Financial Correlations. *International Journal of Theoretical and Applied Finance*. Vol. 3, no. 3, pp. 391–397. ISSN 1793-6322. Available from DOI: 10.1142/s0219024900000255.
- LJUNG, G. M. and BOX, G. E. P., 1978. On a Measure of Lack of Fit in Time Series Models. *Biometrika*. Vol. 65, no. 2, pp. 297–303. ISSN 1464-3510. Available from DOI: <https://doi.org/10.2307/2335207>.
- LOHRE, Harald, NEUGEBAUER, Ulrich and ZIMMER, Carsten, 2012. Diversified Risk Parity Strategies for Equity Portfolio Selection. *Journal of Investing*. Vol. 21, no. 3, pp. 111–128. ISSN 1556-5068. Available from DOI: <https://dx.doi.org/10.2139/ssrn.2049280>.
- MARKOWITZ, Harry, 1952. Portfolio Selection. *The Journal of Finance*. Vol. 7, no. 1, pp. 77–91. ISSN 0022-1082. Available from DOI: <https://doi.org/10.2307/2975974>.
- PARTOVI, M. Hossein and CAPUTO, Michael, 2004. Principal Portfolios: Recasting the Efficient Frontier. *Economics Bulletin*. Vol. 7, no. 3, pp. 1–10. Available also from: <https://EconPapers.repec.org/RePEc:eb1:ecbull:eb-04g00003>.
- SHARPE, William. F., 1964. Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. *The Journal of Finance*. Vol. 19, no. 3, pp. 425–442. ISSN 1540-6261. Available from DOI: <https://doi.org/10.1111/j.1540-6261.1964.tb02865.x>.
- SPEARMAN, Charles, 1904. "General Intelligence," Objectively Determined and Measured. *The American Journal of Psychology*. Vol. 15, no. 2, pp. 201–292. ISSN 00029556. Available from DOI: <https://doi.org/10.2307/1412107>.
- TSAY, Ruey S., 2010. *Analysis of Financial Time Series*. 3rd ed. Wiley. ISBN 0470414359.